Sahaj Bhadja

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Summary

Full-Stack Software Engineer with 2+ years of experience building **Node.js/React applications**, **real-time CRM workflows**, secure integrations, and AI-enabled data products that improve customer conversion.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ consistency across 10K+ submissions through **structured model evaluation**, persistent results, fallback paths, and reviewer controls supporting reproducible testing and responsive user-facing analytical workflows reliably at scale.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **CRM platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with React, Node.js, WebSockets, and Azure SQL across locations reliably.
- Increased lead conversions by 15% through **customer and lead workflow enhancements**, implementing role-based access, real-time updates, notifications, automated tests, monitoring, and reliable Docker/Jenkins deployments across modular services.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web workflows while reducing design errors by 25% through reusable components reliably.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML, CSS
Frontend: React, Next.js, React Native, Responsive Design, Reusable Components, Material UI, Tailwind CSS
Backend & Integrations: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Webhooks, Microservices, API Integration, JWT, Authentication
AI & Data: LLMs, Transformers, PyTorch, Model Evaluation, Feature Engineering, PostgreSQL, MySQL, MongoDB, Redis, AWS RDS, Data Pipelines
Cloud & Quality: AWS, Azure, GCP, Docker, Kubernetes, Git, GitHub, Jenkins, CI/CD, Unit Testing, Integration Testing, PyTest, Jest, Debugging, Monitoring

Projects

Real-Time Lead Conversion Platform - Design Concept
- Designed a **lead-conversion platform** concept joining CRM, web, and product-interaction signals through TypeScript services, PostgreSQL models, scoring pipelines, event-driven updates, customer dashboards, and traceable conversion feedback loops.

AI Sales Agent Integration Hub - Design Concept
- Designed an **AI-agent integration hub** concept with Node.js adapters for CRM and messaging tools, secure token storage, webhook validation, retry queues, conversation state, human handoffs, and audit logs.

FinNexus | Live
- Built a **secure Next.js platform** supporting 50K+ users and 1M+ transactions with Plaid bank linking, token-based access controls, real-time synchronization, advanced filtering, Kubernetes deployment, and MongoDB-backed data access.